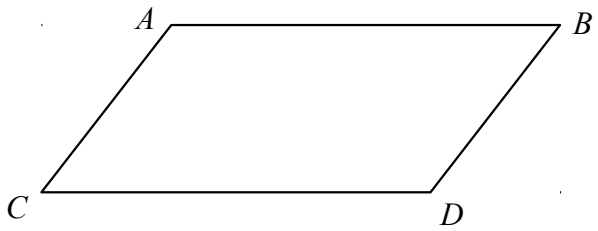
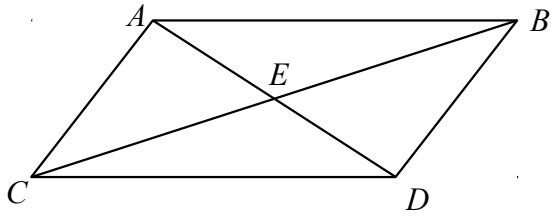
1 *ABCD* is a parallelogram



Prove that triangle *ABC* is congruent to triangle *BCD*.

(3 marks)

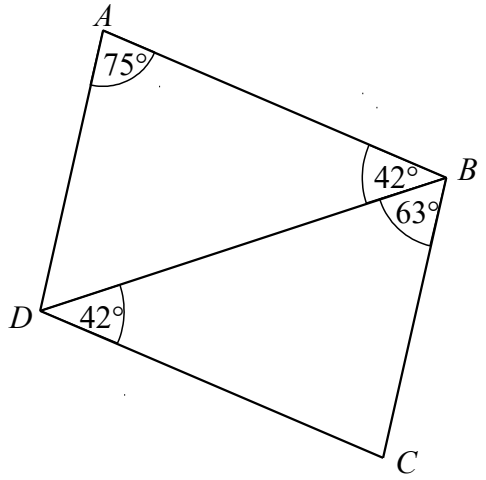
2 *ABCD* is a parallelogram
E is the point where the diagonals *AD* and *BC* meet.



Prove that triangle *ACE* is congruent to triangle *BDE*.

(3 marks)

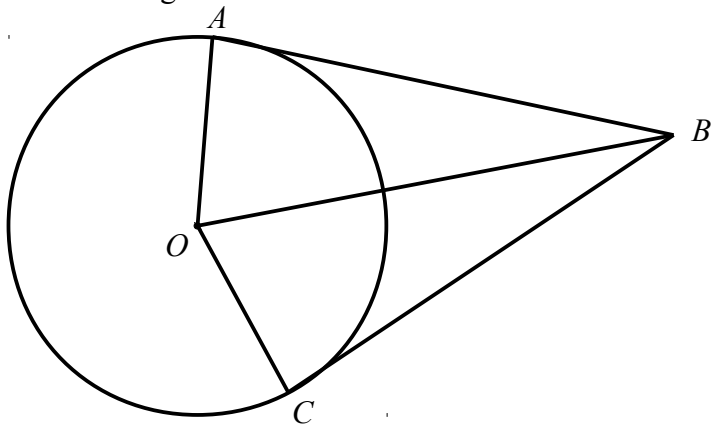
3 The diagram shows two triangles, *ABD* and *BCD*.



Prove that triangle *ABD* is congruent to triangle *BCD*.

(3 marks)

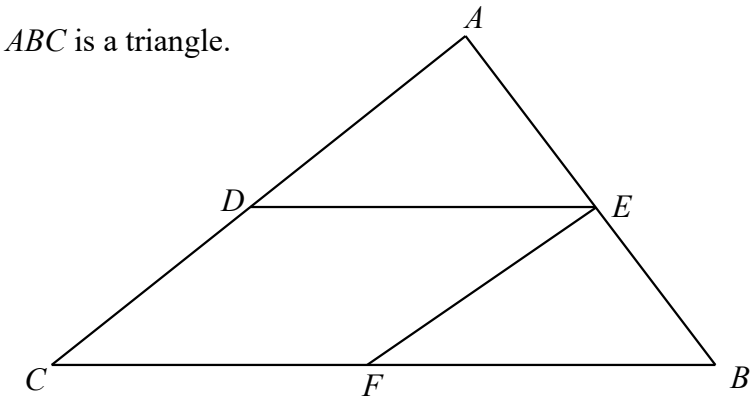
4 *A* and *C* are points on a circle, centre *O*.
AB and *BC* are tangents to the circle.



Prove that triangle *ABO* is congruent to triangle *BCO*.

(4 marks)

5 *ABC* is a triangle.

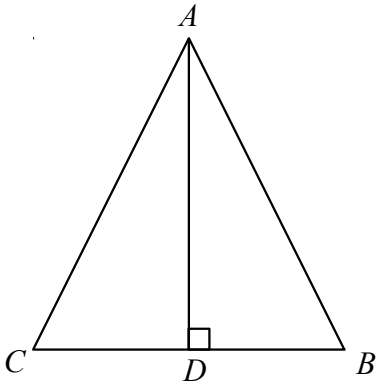


CDEF is a parallelogram such that:
D is the midpoint of *AC*
E is the midpoint of *AB*
F is the midpoint of *BC*

Prove that triangle *ADE* is congruent to triangle *BEF*.

(4 marks)

6 *ABC* is an equilateral triangle.

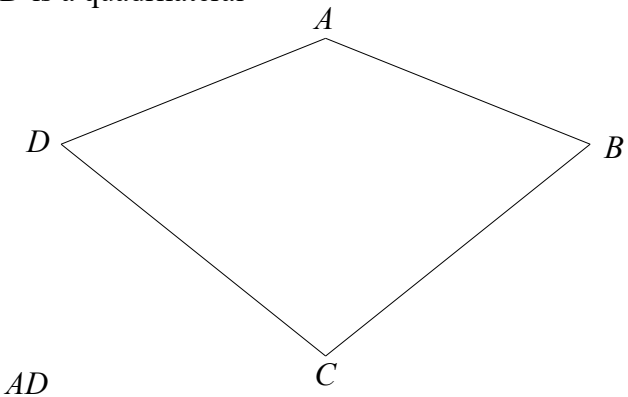


D lies on *BC*
AD is perpendicular to *BC*

Prove that angle *CAD* is equal to angle *BAD*.

(4 marks)

7 *ABCD* is a quadrilateral



AB = *AD*
BC = *CD*

Prove that angle *ABC* is equal to angle *ADC*.

(4 marks)